Maths for ML Handbook

Equations and Formulas used in Machine Learning

1 Linear Equation

1.1 Slope-Intercept Equation

Slope-Intercept Form

$$y = m \cdot x \ + \ c$$

Where,

- m - Slope (angle between straight line and x-axis)
- c - intercept (distance of straight line from the origin)
- x - Independent variable
- y - Dependent variable

Slope

Given two points (x_1, y_1) and (x_2, y_2), formula to calculate Slope m:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Intercept

Given two points (x_1, y_1) and (x_2, y_2), formula to calculate Intercept c:

$$c = y_1 - (m \cdot x_1) \qquad \text{or} \qquad c = y_2 - (m \cdot x_2)$$

1.2 Geometric Relationship b/w lines

Parallel Lines

When given two lines $y = m_1 x + c_1$ and $y = m_2 x + c_2$ are parallel:

$$m_1 = m_2$$

Perpendicular Lines (Orthogonality)

When given two lines $y = m_1 x + c_1$ and $y = m_2 x + c_2$ are perpendicular:

$$m_1 \times m_2 = -1$$

1.3 General Line Equation

Standard Form

General Line Equation (GLE):

$$w_1x + w_2y + w_0 = 0$$

Rearranging above General Line Equation in Slope-Intercept form:

$$y = \left(-\frac{w_1}{w_2}\right)x_1 + \left(-\frac{w_0}{w_2}\right)$$

Where Slope and Intercept can be derived as,

$$m = -\frac{w_1}{w_2} \qquad c = -\frac{w_0}{w_2}$$

Line Equation (2D Hyperplane)

$$w_1x_1 + w_2x_2 + w_0 = 0$$

Plane Equation (3D Hyperplane)

$$w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$$

nD Hyperplane Equation

$$w_1x_1 + w_2x_2 + \ldots + w_nx_n + w_0 = 0$$

2 Vector

2.1 Vector Representation

Mathematical notation of a vector $\vec{x}$ having d dimension containing real numbers:

$$x \in \mathbb{R}^d$$

Where d represents number of elements in the vector.

Row Vector

$$\vec{x} = [x_1\ x_2\ \ldots\ x_d]$$

Column Vector

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

2.2 Types of Distances

1 Euclidean Distance

Formula to calculate Euclidean Distance between two points (x_1, y_1) and (x_2, y_2)

$$d = \sqrt{(y_2 - y_1)^2 + (x_2 - x1)^2}$$

2 Manhattan Distance

Formula to calculate Manhattan Distance between two points (x_1, y_1) and (x_2, y_2)

$$d = |y_2 - y_1| + |x_2 - x_1|$$

2.3 Types of Norms

L2 Norm of a Vector

Formula for L2 Norm or Euclidean Distance of a vector of d dimension:

$$\|\vec{x}\|_2 = \sqrt{{x_1}^2 + {x_2}^2 + {x_3}^2 + \ldots + {x_d}^2}$$

$$= \sqrt{\sum_{i=1}^{d} {x_i}^2}$$

L1 Norm of a Vector

Formula for L1 Norm or Manhattan Distance of a vector of d dimension:

$$\|\vec{x}\|_1 = |x_1| + |x_2| + |x_3| + \ldots + |x_d|$$

$$= \sum_{i=1}^{d} |x_i|$$

3 Matrix Multiplication

3.1 Matrix Multiplication rule

Given two matrices A and B where:

- $A : m \times n$
- $B : n \times k$

Rule

Number of columns in the first matrix should be equal to the number of rows in the second matrix.

$$m \times n \Leftrightarrow n \times k = m \times k$$

The resulting matrix will be of shape: $m \times k$

3.2 Matrix Multiplication in NumPy

Given two NumPy matrices `m1` and `m2`

```
m1 @ m2
# or
np.matmul(m1, m2)
```

4 Dot Product

4.1 Dot Product of Vectors

Formula to calculate dot product of two vectors $\vec{a}$ and $\vec{b}$:

$$\vec{a} \cdot \vec{b} = a^T \cdot b$$

$$= \sum_{i=1}^{d} a_i b_i$$

4.2 Dot Product in NumPy

Given two NumPy vectors `v1` and `v2`

```
np.dot(v1, v2)
```

4.3 Application of Dot Product

1 Angle between Vectors

Formula to calculate angle between two vectors $\vec{a}$ and $\vec{b}$

$$\cos\theta = \frac{a^T \cdot b}{\|a\| \, \|b\|}$$

2 Machine Learning

In ML context, the dot product is used to represent the below "best fit" line or hyper-plane equations:

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

$$\text{or}$$

$$w_1 x_1 + w_2 x_2 + \ldots + w_d x_d + w_0 = 0$$

using $\vec{w}$ and $\vec{x}$ vectors of d dimensions:

$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

as,

$$w^T \cdot x + w_0 = 0$$

Where $w^T \cdot x$ is the dot product between weight vector $\vec{w}$ and feature vector $\vec{x}$.

3 Projection of a Vector

$$\text{Projection of } \vec{x} \text{ on } \vec{y} = \vec{x} \cdot \hat{y}$$

Unit Vector

Formula to calculate unit vector $\widehat{w}$ of a $\vec{w}$ vector:

$$\widehat{w} = \frac{\vec{w}}{\|w\|}$$

Formula to calculate projection of $\vec{x}$ on $\vec{y}$:

$$\text{Projection of } \vec{x} \text{ on } \vec{y} = \frac{\vec{x} \cdot \vec{y}}{\|y\|}$$

5 Shifting Lines

5.1 One Direction

Given line equation:

$$w_1x_1 + w_2x_2 + w_0 = 0$$

Shifting lines in one direction (x-axis or y-axis) by a units:

$$\begin{array}{rrrcrcl}
\text{x-axis}| & \text{Right} \rightarrow & w_1(x_1 - a) & + & w_2x_2 & + & w_0 = 0 \\
\text{x-axis}| & \text{Left} \leftarrow & w_1(x_1 + a) & + & w_2x_2 & + & w_0 = 0 \\
\text{y-axis}| & \text{Up} \uparrow & w_1x_1 & + & w_2(x_2 - a) & + & w_0 = 0 \\
\text{y-axis}| & \text{Down} \downarrow & w_1x_1 & + & w_2(x_2 + a) & + & w_0 = 0
\end{array}$$

5.2 Two Directions

Shifting lines in two direction (x-axis and y-axis) simultaneously by a units:

$$w_1(x_1 - a) + w_2(x_2 - a) + w_0 = 0$$

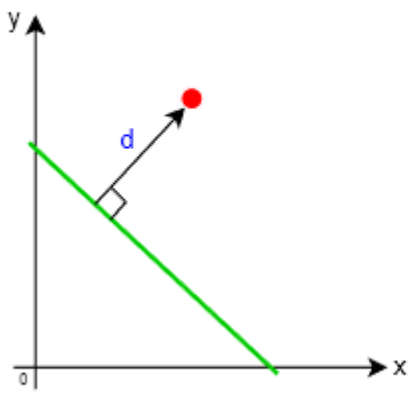
Solving above equation gives:

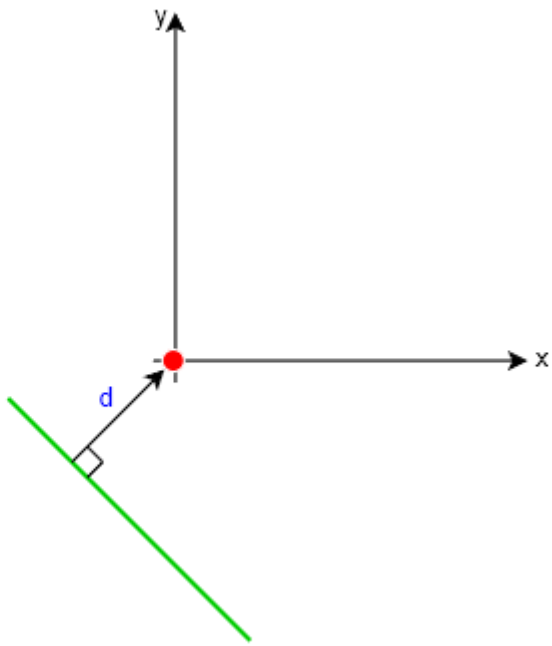
$$w_1x_1 + w_2x_2 + w_0' = 0$$

Where w_0' is:

$$w_0' = w_0 - a(w1 + w2)$$

5.3 Usecase





6 Distances

6.1 Distance Between Line and Origin

Formula to calculate the distance between line (Decision Boundary) and origin using line's Normal vector $\vec{w}$

$$d = \frac{-w_0}{||w||}$$

6.2 Distance Between Point and Line

Formula to calculate the distance between a point x_0 and line (Decision Boundary) using line's Normal vector $\vec{w}$

$$d = \frac{w^T x_0 + w_0}{||w||}$$

6.3 Distance Between Parallel Lines

Same Normal Vector

Formula to calculate the distance between two parallel lines with Normal vector w having same magnitude:

$$d = \frac{w_0^{(1)} \; - \; w_0^{(2)}}{||w||}$$

Different Normal Vector

Formula to calculate the distance between two parallel lines with Normal vectors of different magnitudes where $w^{(1)} = \lambda \cdot w^{(2)}$

$$d = \frac{w_0^{(1)}}{||w^{(1)}||} \; - \; \frac{w_0^{(2)}}{||w^{(2)}||}$$

7 Binary Classification

7.1 Half-spaces

Positive Half-space

In Positive Half-space all data-points have **distance greater than zero** because,

$$w_1x_1 + w_2x_2 + w_0 > 0$$

Negative Half-space

In Negative Half-space all data-points have **distance less than zero** because,

$$w_1x_1 + w_2x_2 + w_0 < 0$$

Decision Boundary

On Decision Boundary all data-points have **distance equal to zero** because,

$$w_1x_1 + w_2x_2 + w_0 = 0$$

7.2 Mathematical Notation

Dataset

A labeled dataset D is represented as:

$$D := \left\{ \left(x_i, y_i\right) \; ; \; x \in \mathbb{R}^d \; ; \; y \in \{-1, +1\} \right\}_{i=1}^{n}$$

Where,

- D is Dataset
- x are features
- y are labels
- n is the size of the dataset

Label

Actual Label y can take values:

$$y \begin{cases} +1 & \text{in +ve Half-Space} \\ \\ -1 & \text{in -ve Half-Space} \end{cases}$$

Classifier

Mapping function for Binary Classification:

$$\begin{aligned}\hat{y} = f(x) &= w_1x_1 + w_2x_2 + \ldots + w_dx_d + w_0 \\ &\text{or} \\ &= w^Tx + w_0\end{aligned}$$

Predicted label $\hat{y}$ can take values:

$$\hat{y} \begin{cases} +1 & \text{if } \left(w^Tx + w_0\right) \geq 0 \\ \\ -1 & \text{if } \left(w^Tx + w_0\right) < 0 \end{cases}$$

7.3 Gain Function

Equation of Gain Function for Binary Classification:

$$g(D, \vec{w}, w_0) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{w^Tx^{(i)} \ + \ w_0}{\|w\|}\right) \cdot y^{(i)}$$

7.4 Loss Function

$$\text{Loss Function} = -\text{ Gain Function}$$

Equation of Loss Function for Binary Classification:

$$l(D, \vec{w}, w_0) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{w^Tx^{(i)} \ + \ w_0}{\|w\|}\right) \cdot y^{(i)}$$

7.5 Perceptron Learning Algorithm

Weight Updation Logic

Logic in Perceptron Learning Algorithm to update Weights for miss-classified data-point:

$$\text{if } \ \text{sign}\left(\frac{w^Tx^{(i)} + w_0}{\|w\|}\right) \ \neq \ \text{sign}\left(y^{(i)}\right) :$$

$$\begin{aligned} w &= w + x^{(i)} \cdot y^{(i)} \\ w_0 &= w_0 + y^{(i)} \end{aligned}$$

8 Optimization

8.1 Optimization of Gain Function

Equation for Optimization of Gain Function for Binary Classification:

$$w^*, w_0^* = \arg\max_{w, w_0} \quad g(D, \vec{w}, w_0)$$

$$= \arg\max_{w, w_0} \frac{1}{n} \sum_{i=1}^{n} \left(\frac{w^T x^{(i)} \ + \ w_0}{\|w\|} \right) \cdot y^{(i)}$$

Goal

Maximize Gain function.

8.2 Optimization of Loss Function

Equation for Optimization of Loss Function for Binary Classification:

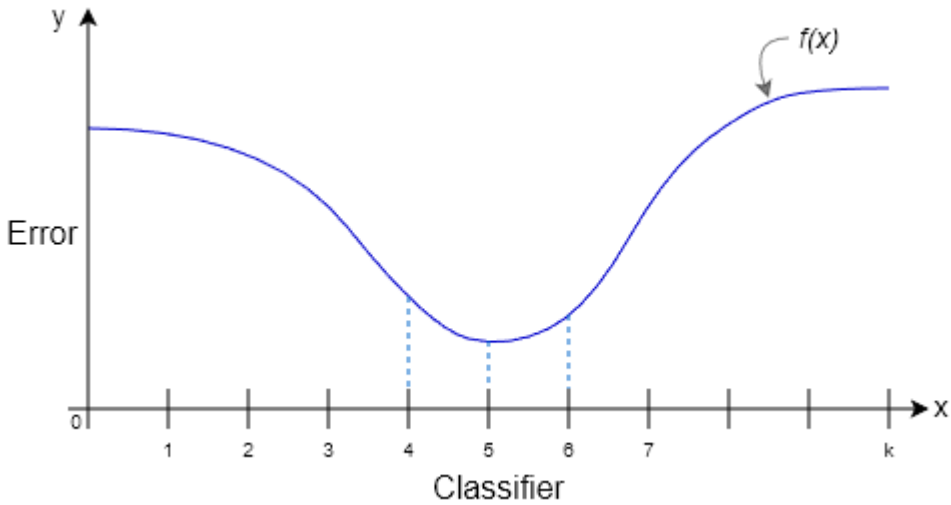
$$w^*, w_0^* = \arg\min_{w, w_0} \quad l(D, \vec{w}, w_0)$$

$$= \arg\min_{w, w_0} \frac{1}{n} \sum_{i=1}^{n} \left(\frac{w^T x^{(i)} \ + \ w_0}{\|w\|} \right) \cdot y^{(i)}$$

Goal

Minimize Loss function.

ML Context



Error vs Classifier is represented as a function $f(x)$ which needs to be minimized.

9 Functions

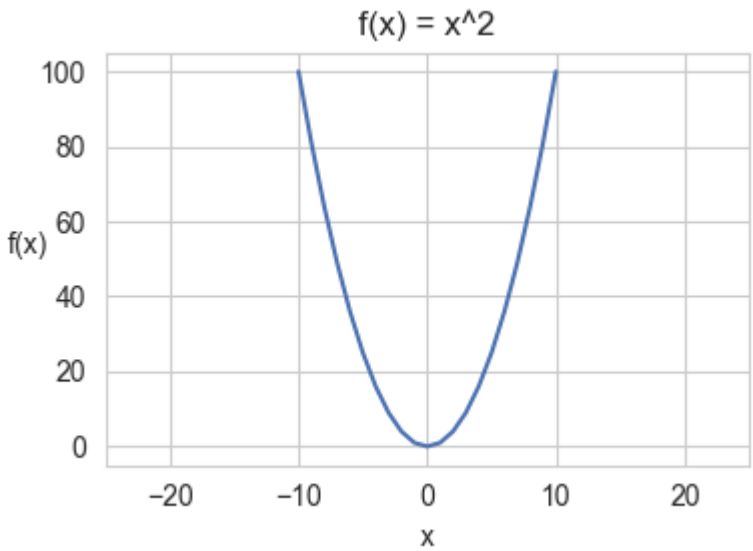
9.1 Continuous Function

Definition

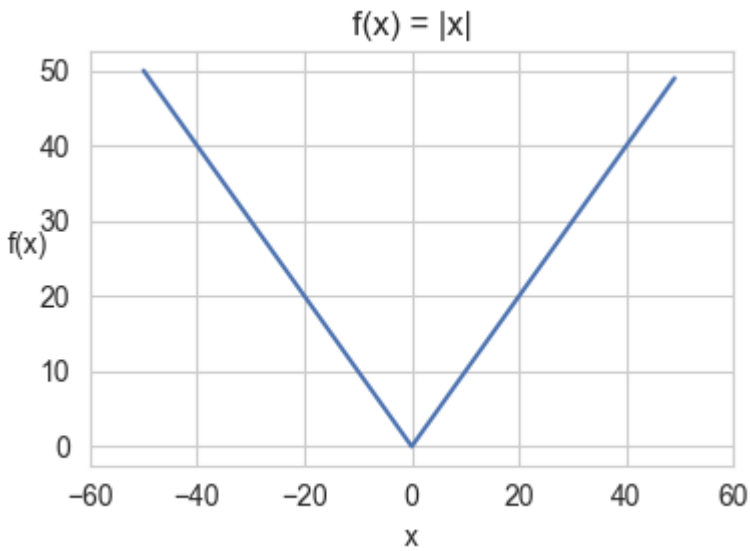
Criteria for a Continuous function:

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$$

Example #1



Example #2



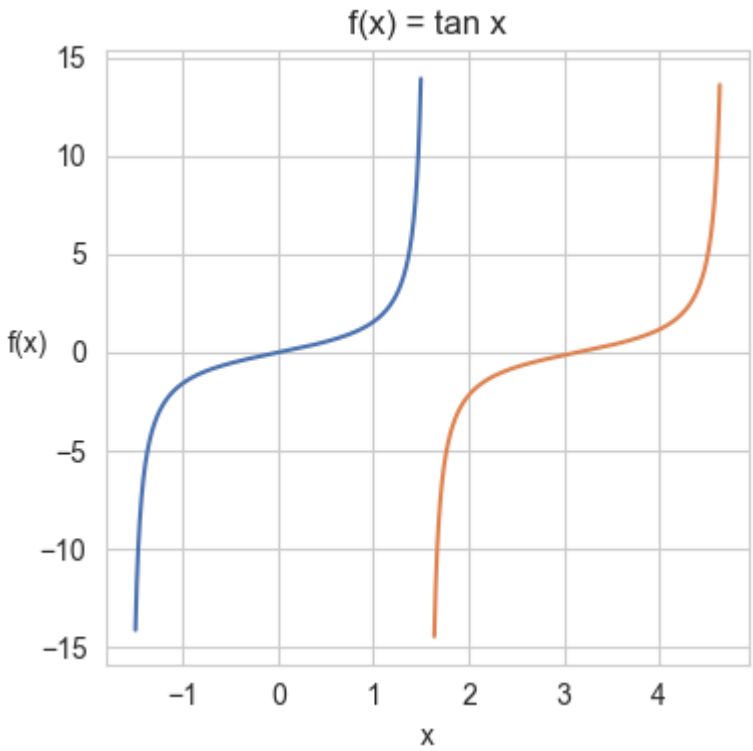
9.2 Discontinuous Function

Definition

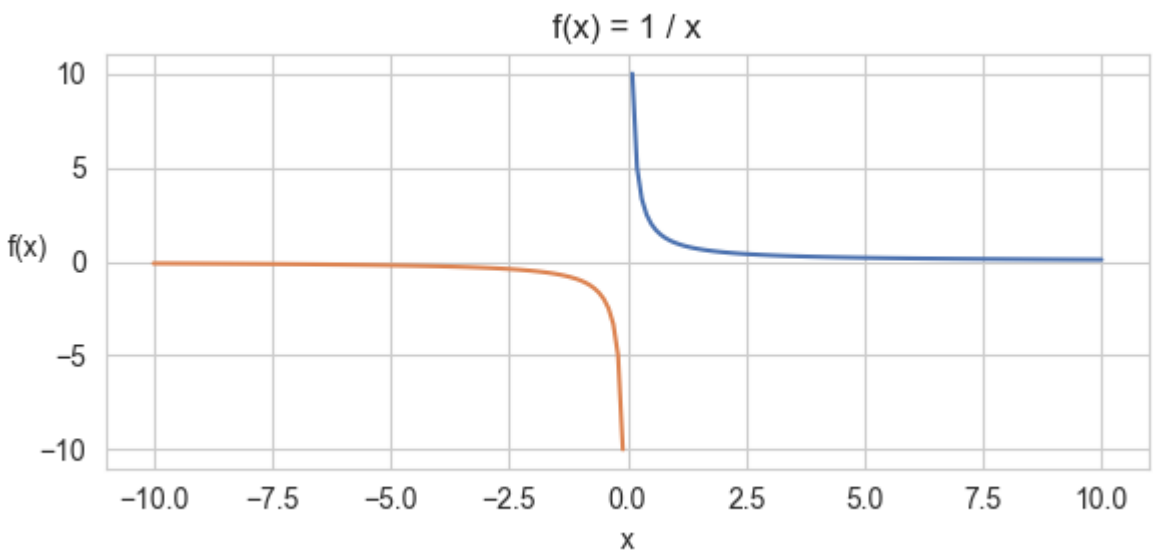
Criteria for a Discontinuous function:

$$\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$$

Example #1



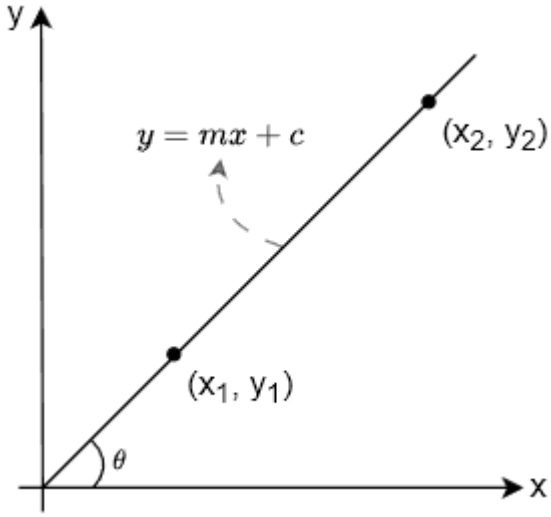
Example #2



10 Differentiation

10.1 Secant

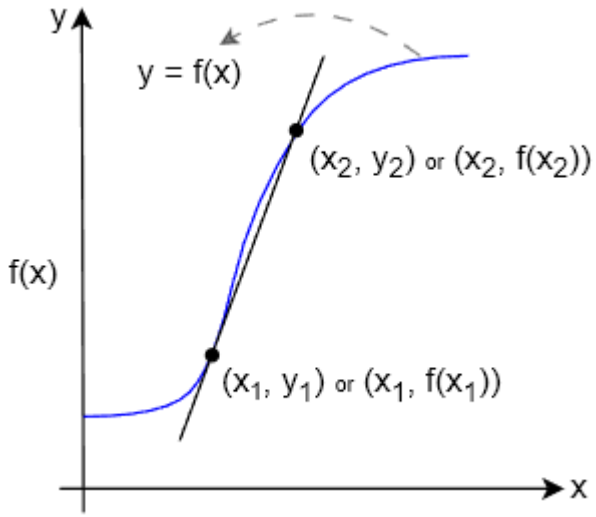
Slope of Line



Formula to calculate slope of a line intersecting two points (x_1, y_1) and (x_2, y_2)

$$m = \tan\theta = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope of Secant Line

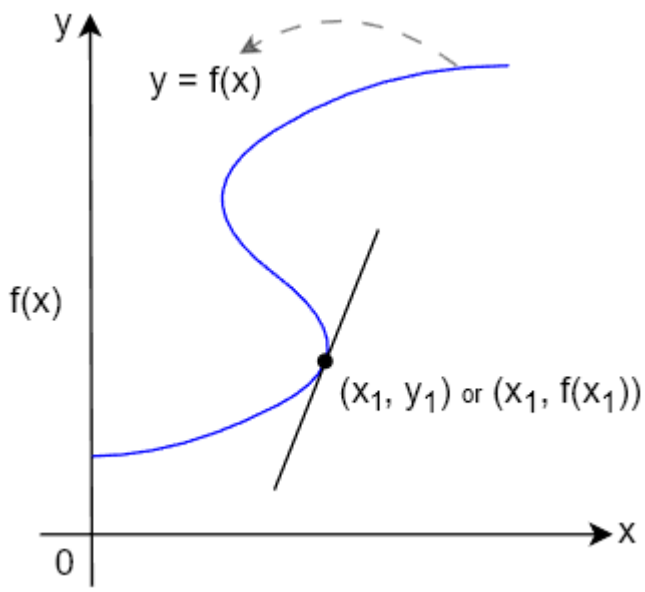


Formula for slope of a secant line intersecting a curve at two points (x_1, y_1) and (x_2, y_2)

$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

10.2 Tangent

Slope of Tangent Line (Derivative)



Formula for slope of a tangent line intersecting a curve at one single point (x, y)

$$f'(x) = \frac{d}{dx} f(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

10.3 Derivatives Examples

Commonly used Derivatives:

$$(1) \quad \frac{d}{dx} x^n = n x^{n-1}$$

$$(2) \quad \frac{d}{dx} \log(x) = \frac{1}{x}$$

$$(3) \quad \frac{d}{dx} e^x = e^x$$

$$(4) \quad \frac{d}{dx} c = 0$$

$$(5) \quad \frac{d}{dx} \sin(x) = \cos(x)$$

$$(6) \quad \frac{d}{dx} \cos(x) = -\sin(x)$$

$$(7) \quad \frac{d}{dx} \tan(x) = \sec^2(x)$$

$$(8) \quad \frac{d}{dx} b^x = b^x \log(b)$$